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Estimating Lateral Detectability of Eutectic Interfaces within Europa's Ice Shell

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Summary: Impurity mixtures within Europa's ice shell will generate melt below their eutectic temperature, which will be detectable by orbital radar sounders. We estimate the lateral detectability of compositionally-relevant eutectic interfaces at various stages of convective vigor.

Abstract

Many geophysical characteristics of the Galilean moon Europa remain poorly constrained, including the thickness of its icy shell (1-90 km). The radar sounding instruments on NASA's Europa Clipper (REASON) and ESA's JUICE (RIME) missions aim to detect the ice-ocean interface, requiring signal penetration through the entire ice shell. The location of sub-surface liquid brine pockets may provide an additional source of constraint. If the eutectic temperature of a non-pure water ice mixture is known, one can study the depth to this eutectic as it's own radar-sensitive interface. This study dynamically simulates Europa's ice shell in two dimensions under various convective regimes, varying ice shell thickness between 20 and 60 km, and reference viscosity over several orders of magnitude. We record the isothermal depths of two compositionally relevant eutectic interfaces (237.15 K and 252.05 K) for each simulation, and then compute lateral detectability based on temperature-dependent signal penetration modeling. We note that cases with very strong, stable convective cells significantly reduce lateral detectability up to 34.1% along our simulated domain. However, we find that all eutectic interfaces are at least locally detectable.

1 Introduction

The Galilean moon Europa is a target of both NASA's Europa Clipper and ESA's JUICE Mission (Pappalardo et al. 2024; Grasset et al. 2013) due in part to its high astrobiological potential. Over two decades ago, NASA's Galileo mission not only revealed the presence of surface salts suggesting a geologically active ice shell (Becker et al. 2024; Carlson et al. 2009), but also an implied salty internal ocean from the detection of an induced magnetic field by the mission's magnetometer (Kivelson et al. 2000). Among the scientific goals of the modern missions to the Galilean moons is the characterization of Europa's ice shell. Radar instruments provide the most direct measurements of sub-surface ice shell investigations. Radar sounding instruments have been deployed on Earth to study the polar ice sheets (Schroeder et al. 2020) and on Mars to study the planet's polar caps (Plaut et al. 2007; Phillips et al. 2008). JUICE and Europa Clipper will each carry radar sounders as part of their instrument suite, allowing for the first direct sub-surface investigations of Galilean icy moons (Bruzzone et al. 2013; Blankenship et al. 2024).

Current constrains on Europa's ice shell range from less than 1 to 90 km based on mechanical, thermal, and impact models (Billings and Kattenhorn 2005; Vilella et al. 2020). Due to the strong reflectance of ice phase transitions at radar frequencies (Dowdeswell and Evans 2004), orbital radar sounders have the potential to directly measure the ice shell thickness if return signals reach the sub-surface ocean. Previous one-dimensional modeling efforts evaluated ice-ocean interface detectability by using thermal profiles to estimate two-way signal attenuation through Europa's ice shell (Chyba, Ostro, and Edwards 1998; Moore 2000; Blankenship et al. 2009; Di Paolo et al. 2017). These studies found that direct sub-surface ocean detection is possible for conductive ice shell thicknesses below 15 km, and decreases by a few kilometers for ice shells that transport heat via convection. While the current convective regime of Europa is unknown, the pits, uplifts, and chaos regions found on the surface suggest convection-driven resurfacing (Pappalardo et al. 1998), which in combination with the large variations in ice shell thickness complicate direct ice-ocean interface detection.

Shallow brines could provide an alternative to ice shell characterization. Melt begins to form within the ice shell if the internal temperature exceeds the eutectic temperature of some impurity mixture. The dielectric transition from the onset of melting will act as a strong signal reflector, providing a laterally consistent return signal from the eutectic isotherm. These isothermal depths will be easier to detect, as their shallower depths would accumulate less attenuation from a radar echo compared to a signal which reached the ice-ocean interface. Recent radar attenuation studies of Enceladus's ice shell investigated the detectability of ammonia and ammonium brines based on their eutectic temperatures, and found the NH_3 eutectic (175.45 K) should be globally detectable regardless of chlorinity concentration or the insulating effect of a porous surface layer (Souček et al. 2023; Byrne et al. 2024). As part of their investigation into how radar attenuation is modified by two-dimensional variations of Europa's thermal structure, Kalousová, Schroeder, and Soderlund (2017) investigated the detectability of a eutectic isotherm at 237 K based on an ocean composition model for Europa which thermodynamically simulates the salt dehydration equilibria, chemical equilibria, and aqueous chemical changes due to sequential salt precipitation of the $Mg - Ca - Na - K - Cl - S - H_2O$ system (Zolotov and Shock 2001). They found that for all investigated ice shells (5-30 km), this particular eutectic interface remained well below the estimated maximum two-way signal attenuation of 100 dB. However, recent estimates from the Juno spacecraft's Microwave Radiometer (Levin et al. 2025) and modeling of multi-basin formation of large surface impactors (Wakita et al. 2024) place a lower bound on Europa's ice shell thickness in the upper tens of kilometers, inviting a reevaluation of eutectic interface detectability within larger ice shells.

This study builds on the work of Kalousová, Schroeder, and Soderlund (2017) by investigating the detectability of eutectic interfaces with a wider range of viscosities ($10^{12} - 10^{15}\ Pa \cdot s$) and thicker ice shells (20-60 km). While previous studies of eutectic interfaces evaluate detection as a binary determination, our study determines interface detectability laterally by comparing isothermal depths to the 100 dB two-way attenuation interface for every radial column in a simulated domain. As the

geometry of the eutectic interface will be heavily modified by the convective state of the ice shell, this parameter study covers all possible convective regimes. We also investigate the effect of a more realistic ice composite rheology, temperature-dependent thermal expansivity and conductivity, and tidal dissipation. Section 2 describes how the ice shell is numerically simulated, how attenuation is calculated, and how we collect and analyze eutectic interfaces. In Section 3 we describe the collective thermal state of all simulations and compare lateral detectability. Section 4 provides relevant context to the results, and the implications of our study are discussed in Section 5.

2 Methods

2.1 Thermal Convection Model

We investigate the relationship between eutectic interfaces and convective vigor, using the finite-volume numerical simulation code GAIA-V2 (Hüttig, Tosi, and Moore 2013). These models solve the equations governing solid-state thermal convection, and employ viscosity ranges relevant to water ice. We use a 2D cylindrical geometry (Jarvis 1993), where the ice shell is divided between 79 and 270 radial shells, each containing between 772 and 1750 cells which are uniformly distributed. The number of radial and lateral layers was chosen based on ice shell thickness. All simulations are given the free slip boundary condition, although the top layer naturally evolves into a no-slip boundary due to the strong temperature-dependence of viscosity. Our models use the Extended Boussinesq Approximation (EBA) to account for adiabatic heating and cooling (Schubert, Turcotte, and Olson 2001). Assuming an infinite Prandtl number, the non-dimensional form of the conservation equations (mass, momentum, energy) reads as follows:

$$\nabla \cdot \vec{v'} = 0, \tag{1}$$

$$\nabla p' - \nabla \cdot \left(\eta' \left(\nabla \vec{v'} + (\nabla \vec{v'})^T \right) \right) = Ra\, \alpha\, T'\, \vec{e}_r, \tag{2}$$

$$\frac{\partial T'}{\partial t'} + \vec{v'} \cdot \nabla T' - Di\, \alpha' \left(T' + T'_{surf}\right) \vec{v'}_r - \frac{Di}{Ra} \Phi = \nabla \cdot (k' \nabla T') + H. \tag{3}$$

Here $\vec{v}$ is velocity, v_r is the radial velocity component, η is viscosity, T is temperature, T_{surf} is the surface temperature, p is dynamic pressure, k is thermal conductivity, H is volumetric heating, and t is time. Φ, or the viscous dissipation parameter, can be expressed as $\tau : \dot{\epsilon}/2$, where $\dot{\epsilon}$ is the strain rate tensor, and τ is the deviatoric stress tensor.

It is often helpful, when describing aspects of convective systems, to represent various behaviors of interest by characteristic non-dimensional numbers. The Rayleigh number Ra characterizes the vigor of convection in a system and is defined as follows:

$$Ra = \frac{\rho_{ref} g \alpha_{ref} \Delta T D^3}{\eta_{ref} \kappa_{ref}}. \tag{4}$$

With ρ_{ref} as density, g as the acceleration due to gravity, and α_{ref} as thermal expansivity. The reference label under each parameter indicates that the parameter was obtained in laboratory experiments, under reference conditions. Another dimensionless quantity is the dissipation number Di, which characterizes the adiabatic heating and cooling strength of a convective system. Di is defined as follows:

$$Di = \frac{\alpha_{ref} g D}{c_p}, \tag{5}$$

where c_p is the specific heat capacity of water ice.

Equations 1-3 are non-dimensionalized using ice shell thickness D, the temperature difference between the surface ΔT, a reference viscosity η_{ref}, and thermal diffusivity κ_{ref}. Time is non-dimensionalized based on the thermal diffusion timescale:

$$t' = t\frac{\kappa}{D^2}, \tag{6}$$

Temperature T by the contrast between the surface temperature and the ice-ocean interface temperature T_b:

$$T' = \frac{T - T_{surf}}{\Delta T}, \tag{7}$$

viscosity with a reference viscosity η_{ref}, taken as an average viscosity at the ice-ocean interface:

$$\eta' = \frac{\eta}{\eta_{ref}}, \tag{8}$$

pressure using the thermal diffusivity time scale and the same reference viscosity η_{ref}:

$$p' = \frac{pD^2}{\eta_{ref}\kappa}, \tag{9}$$

and depth by the ice shell thickness:

$$z' = \frac{z}{D} \tag{10}$$

Table 1 contains all constants used in the above equations, and Table 2 contains all parameters which vary with ice shell thickness. Values selected for this study are representative of Europa's equatorial region.

Table 1: Costants Used Across Thermal Evolution Models

Parameter	Symbol	Value	Unit
[a]Density	ρ_{ref}	920	$kg \cdot m^{-3}$
[b]Gravitational Acceleration	g	1.315	$m \cdot s^{-2}$
[c]Surface Temperature	T_{surf}	110	K
[b]Specific Heat Capacity	c_p	2090	$J \cdot kg^{-1} \cdot K^{-1}$

[a] Hobbs (1974)
[b] Schubert, Sohl, and Hussmann (2009)
[c] Ojakangas (1989)

Each simulation was run until it reached a quasi-steady state, but for a minimum of at least 15 Myr dimensional time. The quasi-steady state is determined by the root mean squared velocity (v_{rms}) at each timestep oscillating around a single mean value (Supplemental Figure 1). Simply using any field from the final timestep of a simulation may give misleading results if the simulation, for example, was immediately following some form of convective plume collapse. Therefore, to remove these potential biases from our study, we opted to time-average all spatial fields over the final 10% of all simulation time steps after they have reached a statistical steady-state.

2.2 Ice Shell Rheology

The viscosity of water ice is known to be controlled by at least four deformation mechanisms: diffusion, dislocation, grain-boundary sliding, and basal slip (Goldsby and Kohlstedt 2001). Under the

Table 2: Ice Shell Thickness Varying Parameters for Thermal Evolution Model

[a] D (km)	[a] T_b (K)	[b] α_{ref} (K^{-1})	[c] k_{ref} (W m^{-1} K^{-1})	κ_{ref} (m^2 s^{-1})
20	271.2	$1.544 \cdot 10^{-4}$	2.400	$1.248 \cdot 10^{-6}$
30	270.2	$1.518 \cdot 10^{-4}$	2.409	$1.252 \cdot 10^{-6}$
40	269.2	$1.492 \cdot 10^{-4}$	2.418	$1.257 \cdot 10^{-6}$
50	268.1	$1.465 \cdot 10^{-4}$	2.428	$1.262 \cdot 10^{-6}$
60	267.0	$1.438 \cdot 10^{-4}$	2.439	$1.268 \cdot 10^{-6}$

[a] Schubert, Sohl, and Hussmann (2009)
[b] Feistel and Wagner (2006)
[c] Petrenko and Whitworth (1999)

assumption of isotropy, η can be expressed via the "iso-strain model" (Schultz et al. 2022; Hirth and Kohlstedt 2003):

$$\eta = \frac{d^{\frac{m_i}{n_i}} \dot{\epsilon}^{\frac{n_i-1}{n_i}}}{2A_i} \exp\left(\frac{E_i + V_i P_{\text{ref}}}{n_i R T_{\text{ref}}}\right), \tag{11}$$

where d is grain size, R is the universal gas constant, and P_{ref} is the reference pressure at the ice-ocean interface. All $_i$ components are deformation-mechanism specific parameters determined experimentally. A_i is a prefactor, E_i is the activation energy, and V_i is the activation volume. m_i and n_i are the grain size and stress exponents, respectively. Mechanism-specific parameters can be found in table 3. We follow Schubert, Turcotte, and Olson (2001) non-dimensionalization of η, involving a reference viscosity η_{ref}:

Table 3: Rheological Parameters Per Deformation Mechanism

Mechanism	n	m	A_{cold} ($m^m Pa^{-n} s^{-1}$)	E_{cold} (kJ mol^{-1})	A_{warm} ($m^m Pa^{-n} s^{-1}$)	E_{warm} (kJ mol^{-1})
Diffusion creep	1.0	2.0	3.3×10^{-10}	59.4	3.3×10^{-10}	59.4
Dislocation creep	4.0	0.0	4.0×10^{-19}	60.0	6.0×10^{4}	180.0
Grain-boundary sliding	1.8	1.4	6.2×10^{-14}	49.0	4.8×10^{15}	192.0
Basal slip	2.4	0.0	2.2×10^{-7}	60.0	2.2×10^{-7}	60.0

* Values from Goldsby and Kohlstedt (2001) and Durham, Stern, and Kirby (2001). A and E differ above/below the transition temperatures of 258 K (dislocation creep) and 255 K (grain-boundary sliding), denoted warm/cold.

$$\eta' = \frac{G^{\frac{m_i}{n_i}} \dot{\epsilon}'^{\frac{n_i-1}{n_i}} \left(\frac{\kappa}{D^2}\right)^{\frac{n_i-1}{n_i}}}{2A_i \eta_{ref}} \exp\left(\frac{E_i' + z' V_i'}{n_i \left(T' + T'_{surf}\right)}\right) \tag{12}$$

Following the composite rheology framework introduced by Goldsby and Kohlstedt (2001), the effective viscosity η_{eff} is calculated from contributions of each mechanism-specific viscosity function in the form of a harmonic average (Harel et al. 2020):

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} + \frac{1}{\eta_{gbs} + \eta_{bs}} \tag{13}$$

For the sake of simplicity, previous studies neglect the effect of pressure and implement temperature-dependent viscosity for a purely diffusion controlled rheology (Kirk and Stevenson 1983; Showman

and Han 2004; Tobie et al. 2005; Běhounková et al. 2012; Kalousová, Schroeder, and Soderlund 2017). The dimensional form of this Arrhenius relation can be written as:

$$\eta = \eta_{ref} \exp\left(\frac{E_a}{RT_b}\left(\frac{T_b}{T} - 1\right)\right) \tag{14}$$

We generated a total of 45 simulations for this study, which we divide into two sets. The first set of 20 baseline models span the largest possible parameter space in terms ice shell thickness and reference viscosity (both first-order controls on convective vigor), but maintain simplistic rheological and thermal parameters, including the simple formulation of viscosity demonstrated in equation 14. The second set of 25 simulations individually enable depth-dependent thermal conductivity and expansivity, tidal heating, and the composite rheology (equations 11 and 12) for all explored ice shell thicknesses but for a single reference visocisty of $10^{13}\ Pa \cdot s$. This division is done to illustrate the separate effects of more realistic geophysical behaviors on the lateral detectability of eutectic interfaces. Details of which behaviors are enabled within which set of simulations can be found in the next two subsections.

2.3 Baseline Models

All baseline models use the simple temperature-dependent viscosity formulation given in equation 14. We then treat reference viscosity η_{ref} as a free parameter, varying it over several orders of magnitude (10^{12} to 10^{15} Pa s). This range is consistent with values found in literature (Montagnat and Duval 2000; Barr, Pappalardo, and Zhong 2004; Howell 2021). All diffusion-specific parameters are taken from Table 1 in Kalousová, Schroeder, and Soderlund (2017). While recent literature values place consistent lower bounds of Europa's ice shell at 20 km (Wakita et al. 2024; Levin et al. 2025), an upper bounds seems much less constrained (Howell 2021). Due to the weak upper limit in ice shell thickness, and the effect that D has on the convective vigor of a system (4), we vary ice shell thickness between 20 and 60 km in steps of 10 km. All baseline simulations additionally neglect the effect of tidal heating, and maintain a constant depth-wise thermal conductivity and expansivity based on reference conditions at the ice-ocean interface (see Table 2).

2.4 Additional Parameter Variation

In addition to the baseline set of 20 simulations, we explore the effects of a complex rheology, by including the contributions from diffusion creep, dislocation creep, grain-boundary sliding, and basal slip. Prescribing a grain size for the complex rheology is complicated by lack of extraterrestrial ice grain size measurements. Goldsby and Kohlstedt (2001) studied terrestrial ice grain sizes at pressures relevant for polycrystalline ice, between 0.1 mm and and 10s of mm. With diffusion being the dominant creep mechanism in the complex rheology simulations (Supplemental Figure 7), we prescribe a grain size of 0.16 mm based on diffusion reference conditions at the ice-ocean interface assuming a viscosity of 10^{13} Pa s (Supplemental Figure 12).

In another set of simulations we use temperature- and pressure-dependent thermal expansivity $\alpha(T, p)$ and temperature-dependent thermal conductivity $k(T)$. Previous studies have demonstrated the effects of a temperature-dependent thermal conductivity on the detectability of the ice-ocean interface (Kalousová, Schroeder, and Soderlund 2017; Souček et al. 2023). Both Kalousová, Schroeder, and Soderlund (2017) and Souček et al. (2023) consider a simple inverse relationship between thermal conductivity and temperature, based on laboratory measurements of pure water ice (Hobbs 1974; Petrenko and Whitworth 1999). Petrenko and Whitworth (1999) demonstrated that the thermal conductivity for terrestrial polycrystalline ice is well fitted by the following dimensional inverse relation at temperatures above 60 K:

$$k(T) = \frac{651}{T} \tag{15}$$

When this relation is in use, temperature-dependent thermal conductivity values will range from 5.92 $Wm^{-1}K^{-1}$ at the surface to the same values given in Table 2 (i.e., $\sim 2.4\ Wm^{-1}K^{-1}$) at the ice-ocean interface. When considering variable thermal expansivity, we use a fit (Rückriemen-Bez and Plesa 2025) derived from an extensive collection of thermodynamic equilibrium properties of ice 1h (Feistel and Wagner 2006). This fit is composed of the sum of two univariate polynomials ($A(T)$, $B(T)$), whose dimensional coefficients are listed in Supplemental Table 1. While Feistel and Wagner (2006) express expansivity in terms of pressure, Rückriemen-Bez and Plesa (2025) adopt a depth-dependent representation using the $p = \rho g z$ relation. Following this approach, we rewrite the thermal expansivity dimensionally as:

$$\alpha(T, z) = A(T) + B(T) \cdot g \cdot \rho \cdot z \tag{16}$$

Simulations with temperature-pressure dependent thermal expansivity have a value of $3.87 \times 10^{-5}K^{-1}$ at the surface and range from $1.44 \times 10^{-4}K^{-1}$ and $1.54 \times 10^{-4}K^{-1}$ at the ice-ocean interface depending on the ice-shell thickness. Depth-wise profiles for variable thermal expansivity and conductivity are shown in Supplemental Figure 3 and 4. An additional five simulations were run with both $k(T)$ and $\alpha(T, z)$ enabled to observe which effect dominates the thermal structure of the ice shell.

Europa experiences intense tidal forces owing to its orbital resonance with Galileans moons Io and Ganymede, and its interaction with Jupiter's gravitational field. To understand how the thermal structure variations induced by tidal heating on Europa's ice shell effect eutectic interface detectability, we generated an additional five simulations with tidal heating enabled. Under the assumption of a four layer structural model of Europa and an incompressible Maxwell medium, heat from tidal dissipation can be dimensionally parametrized as follows (Tobie, Choblet, and Sotin 2003):

$$H = \frac{2H_{max}}{\frac{\eta}{\eta_{max}} + \frac{\eta_{max}}{\eta}}. \tag{17}$$

Tobie, Choblet, and Sotin (2003) found that tidal dissipation peaks when the viscous relaxation timescale is comparable to the tidal forcing period (or when $\eta_{max} = \mu_E/\omega$). The maximum tidal heating rate H_{max} attains a value of $2.10 \times 10^{-6} Wm^{-3}$ at the equator for a viscosity of $\eta_{max} = 1.5 \times 10^{14}\ Pa \cdot s$.

2.5 Radar Sounding Attenuation

The detectability of any sub-surface radar sounding feature is constrained by the amount of signal degradation, known as two-way attenuation (Blankenship et al. 2009; Bruzzone et al. 2015; Di Paolo et al. 2017). Kalousová, Schroeder, and Soderlund (2017) have estimated a maximum two-way attenuation limit of 100 dB, past which one would not expect any detectable return signal. The one-way englacial attenuation rate A_1 (measured in dB/km) is known to be a function of the electrical conductivity of ice σ (measured in $\mu S/m$) within the HF-VHF frequency range that most planetary and terrestrial sounders operate in. A typical expression of A_1, given by Matsuoka, MacGregor, and Pattyn (2010), is:

$$A_1 \approx 0.914\sigma \tag{18}$$

We follow Corr, Moore, and Nicholls (1993) in assuming the electrical conductivity contribution of ice sheets come from pure water ice ($i = 0$), and impurities such as acids ($i = 1$), and ocean salts ($i = 2$):

$$\sigma = \sum_{i=0}^{2} \sigma_i^0 C_i \exp\left[-\frac{E_i}{k_b}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right] \tag{19}$$

Here σ_i^0 is the molar conductivity, C_i is the molar concentration,k_b is the Boltzmann constant, and E_i is the activation energy. The temperature dependence of electrical conductivity is modeled as

an Arrhenius relationship based on known electrical conductivity values from terrestrial ice samples taken at a reference temperature T_{ref} of 251 K (Moore, Paren, and Oerter 1992). Each summand represents the contribution of a specific compositional component to the overall electrical conductivity. Kalousová, Schroeder, and Soderlund (2017) additionally described two end-member attenuation models to account for plausible attenuation ranges within Europan ice. One where attenuation is calculated only with electrical conductivity contributions from pure water ice (i = 0), which is referred to as the "low loss" model. The other end-member, the so called "high loss" model includes electrical conductivity contributions acids, and sea salts (i = 1, 2). All components used for electrical conductivity calculations are listed in Table 1 of Kalousová, Schroeder, and Soderlund (2017).

Once an attenuation model is selected, we can calculate attenuation as a function of depth, radially, from the surface of our domain to the ice-ocean interface:

$$A_2(d) = \sum_{i=0}^{i_d} 2A_1(d_i)\Delta d, \tag{20}$$

where d_i is the depth to the ith cell in one radial column, $A_1(d_i)$ is the one-way attenuation rate at that depth, and Δd is the thickness of the layer in which A_1 was calculated.

2.6 Eutectic Interfaces

Europa's ice shell is thought to be predominantly composed of water ice, with a fraction of salts and other impurities (Schubert, Sohl, and Hussmann 2009). The structural expectation is that the ice shell thickness will be limited in bulk by the depth at which the melting point of pure water ice (273.15 K) is reached. However, local concentrations of impurities may allow for melt to form at lower temperatures, and hence at shallower depths. Such pockets of shallow melt could be detectable by radar sounding instruments. We investigate the depth of two eutectic interfaces of interest based on previous literature, given that shallow pockets of melt can form at these interfaces and be detectable by radar sounding. The first is the eutectic temperature of a modeled Europan ocean, which is expected to be around 237.15 K (Zolotov and Shock 2001). The second eutectic temperature is based on a mixture of sodium chloride and pure water ice, which is expected to be around 252.05 K (Wolfenbarger et al. 2022). We denote these eutectic temperature using the following nomenclature:

$$T_{e,Z} = 237.15K \tag{21}$$

$$T_{e,N} = 252.05K \tag{22}$$

To simulate the existence of these eutectics, we record the depth within each simulation where the internal temperature first crosses the eutectic temperature, starting from the ice-ocean interface, for every radial column within our computational domain. This record of radial depths is treated as an isotherm, or *eutectic interface*, the characteristics of which are investigated in this study. While previous radar-sounding geodynamic studies of icy moons have focused on the ice-ocean interface detection (Kalousová, Schroeder, and Soderlund 2017; Souček et al. 2023), this study specifically focuses on the lateral detectability of these impurity melt zones, improving the assessment of eutectic interfaces as a scientific focus for planetary radar sounding of icy moons.

3 Results

Since any eutectic interface's geometry will be tied to the thermal structure of the simulation, it is necessary to span all possible convective regimes. Within the reference viscosity range of our study ($10^{12} - 10^{15}\ Pa \cdot s$), we observed three distinct regimes of convective vigor. The distinction between them is based on the thermal structure of the simulation, and the temporal evolution of a

simulation's v_{rms} (Supplemental Figure 1 and 2). The first state describes a thermal system that moves heat purely through conduction. This is characterized by a horizontally uniform depth-wise gradient within the temperature field, and typically referred to as the conductive regime. In planetary geodynamics, the labels for the two nominally referenced convective states are not standardized, but they do have commonly cited distinguishable differences. The first convective state is denoted by rounded, symmetrical upwellings (or plume) structures that take up the bottom half of the domain, with a large conductive lid above. These simulations typically reach a steady state, with a constant v_{rms} after some computation time. Convection within the second state is more vigorous, characterized by frequent time-dependent plume movement. The increased convective strength causes more of the domain to be thermally well-mixed, leading to lower overall conductive lid thicknesses, but with more high-frequency lateral variations. This second state does not reach a true steady-state, rather it reaches a "statistical" or "quasi" steady-state, where v_{rms} oscillates around a mean value. Some authors focus on the time-dependence of the second state, labeling it simply as the "Time-Dependent" regime (e.g. Harel et al. (2020)), while others focus on the vigor of convection, giving a label like the "convective chaotic" regime (Kalousová, Schroeder, and Soderlund 2017). For the description of our results, we opt to use the labels "sluggishly" convective to refer to the first convective regime, and "vigorously" convective to refer to the second.

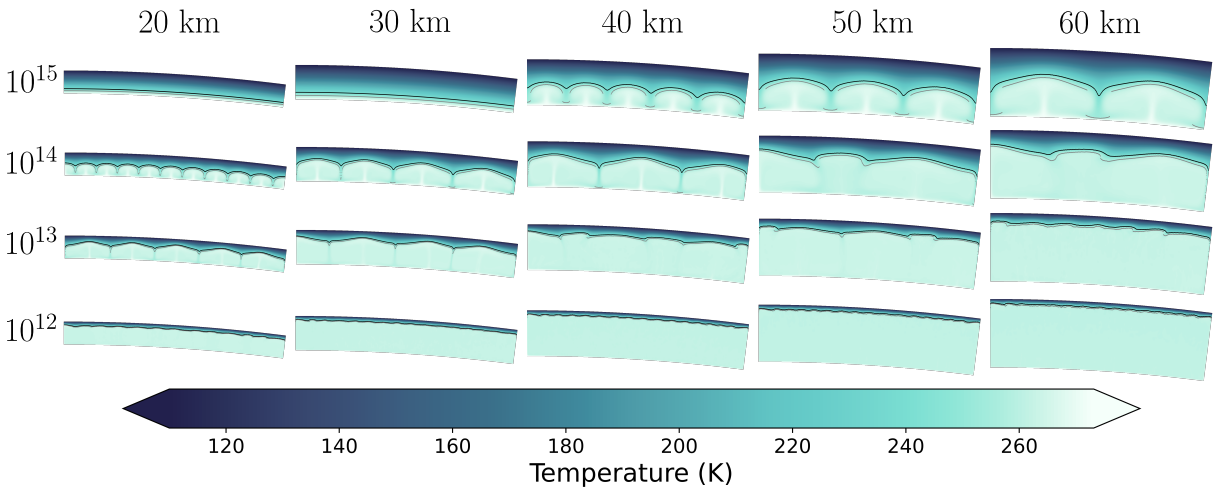


Figure 1: Time-averaged temperature fields after each simulation has reached a quasi-steady state. Fields are organized by ice shell thickness (D) and reference viscosity (η_{ref}). Black and gray lines mark the depth to the 237.15 K and 252.05 K eutectic temperatures.

Figure 1 shows the time-averaged temperature fields of all baseline simulations, after they have run for at least 15 Myr, and have reached either a steady or quasi-steady state. Of our set of 20 baseline simulations, two are conductive, six are sluggishly convective, and 12 vigorously convective. Modifications in reference viscosity (row-wise in Figure 1) reveal a few trends across the baseline simulations. The boundary between the conductive regime and the sluggishly convective regime is crossed once (implying Ra has surpassed the critical Ra number) between the 30 and 40 km ice shell thickness cases with an η_{ref} of 10^{15} $Pa \cdot s$. The boundary between the sluggishly convective and vigorously convective regime is also only crossed once at a reference viscosity of 10^{14} $Pa \cdot s$, between 40 and 50 km. While the two lowest ice shell thickness simulations for the 10^{13} $Pa \cdot s$ cases appear to have stable convective cells, minor plume instabilities are sufficient to make the v_{rms} behavior oscillatory (Supplemental Figure 1). The simulations with the lowest η_{ref} remain vigorously convective across all variations in ice shell thickness. When observing Figure 1 column-wise, only the 20 and 30 km ice shell thickness cases span all three convective regimes, while the higher ice shell thicknesses create such large thermal gradients that they are either sluggishly or vigorously convective.

Additional simulations (described in section 2.4) were ran based on the 10^{13} $Pa \cdot s$ reference

viscosity baseline simulations. The low ice-shell thickness simulations in this set (20 and 30 km) have clearly defined symmetric plumes but still have time-dependent instabilities placing them in the vigorously convective regime. We note that the inclusion of more complex mechanisms within the ice shell will effect eutectic interface geometry and, thus, lead to different thermal structures than the baseline models.

When enabling only variable thermal conductivity ($k(T), \alpha_0$), an addition plume forms, along with a notable increase in conductive lid thickness between 2-3 kilometers across ice shell thicknesses. This general reduction in convective vigor compared to the baseline cases is due to the significant increase in thermal conductivity for the upper portion of the ice shell (Supplemental Figure 3) that promotes heat transport by diffusion. The drop in convective vigor is substantial enough to move the 20 km ice shell case from the vigorously convective to sluggishly convective regime (Supplemental Figure 2). Unlike the simulations with a variable thermal conductivity, simulations that employ a temperature- and depth-dependent thermal expansivity ($k_0, \alpha(T, z)$) have the same number of convective cells compared to the baseline case (20 km, $\eta_{ref} = 10^{13}\ Pa \cdot s$), and have a notably smaller conductive lid by 0.5 - 1 km. The thinner conductive layer reflects an increase in convective vigor compared to the baseline scenarios, additionally supported by the increase in mean v_{rms} across the range of ice shell thicknesses. Indeed, the higher thermal expansivity compared to the baseline cases, as illustrated by the thermal expansivity profile (Supplemental Figure 4) increases the Rayleigh number and hence the convective vigor. When enabling both temperature-dependent thermal conductivity and temperature- and depth-dependent thermal expansivity, the increased near-surface conductivity dominates, leading to an overall decrease in convective vigor. The additional plume re-appears in the 20 km case, and that simulation eventually reaches a convective steady state. The conductive lid is raised by about 1-3 km.

While the 20 km ice shell simulation for a mixed rheology (η_{mix}), but constant thermal conductivity and expansivity, does eventually reach a convective steady state (rather than an oscillating, quasi-steady state), it does not develop an additional plume. There is an overall increase in conductive lid thickness across all five simulations (1-3 km), but slightly less than the simulations with a variable thermal conductivity. The inclusion of tidal heating (H) causes the most significant increase convective vigor, with a mean shrinking of conductive lid thickness of ~ 1 km. While the larger ice shell thickness in the additional simulations (30-60 km) contain more low frequency variations in lateral conductive lid thickness (Supplemental Figure 5), local lid thicknesses for tidally heated simulations contain notably more high-frequency variations.

These modifications of convective vigor, especially variations in the conductive lid thickness, are easily observable in lateral variations of the eutectic interface geometry. Figure 2 visualizes the coldest eutectic interface (237.15 K) for end-member ice shell thicknesses, using a different color for each reference viscosity. The convective thermal structure is clearly reflected in the interface geometry. Within the collection of 20 km ice shell thicknesses, we observe a laterally constant eutectic interface for the one conductive simulation ($\log(\eta_{ref}) = 15$), rounded symmetrical plume heads clearly visible along the interface for sluggishly convective simulations ($\log(\eta_{ref}) = 14$), and the low amplitude, high-frequency variations that come from vigorous convection in the simulations with a low reference viscosity ($\log(\eta_{ref}) = 12$).

The variations in eutectic interface geometry shown in figure 2 reflects the expected positive trend of increased convective vigor with increased ice shell thicknesses, and decreased convective vigor with increasing reference viscosity. Supplemental Figures 10 and 11 show all eutectic interfaces for each eutectic temperature and ice shell thickness of all simulations in this study. Due to the increased eutectic temperature of the sodium chloride end-member (252.05 K), the 237.15 K eutectic interface lies at a shallower average depth. Eutectic interfaces within sluggishly convective simulations are able to span 10s of km in some simulations depending on how deep cold downwellings penetrate into the ice shell.

While non-baseline eutectic interfaces predominantly maintain the same observable trends as their $10^{13}\ Pa \cdot s$ baseline counterpart, there are a few additional relations that are worth noting.

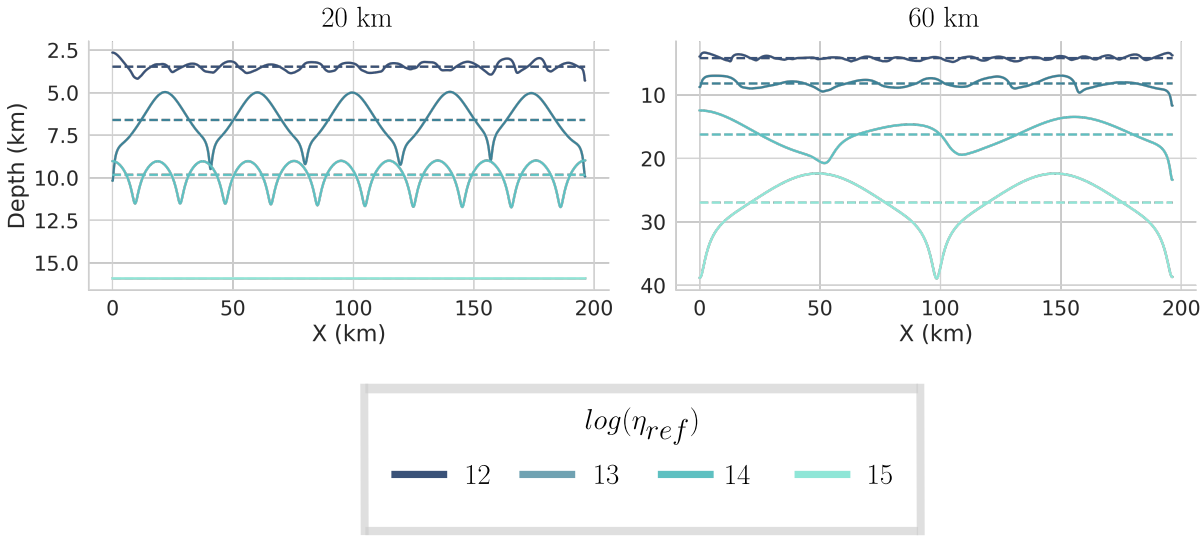


Figure 2: All 237.15 K eutectic interfaces plotted for all variations in η_{ref} (color) for end-member ice shell thicknesses (20 and 60 km). Dashed line represents the mean eutectic depth for a given isotherm.

When comparing eutectic interface depths from non-baseline simulations, interfaces gathered from simulations with variable thermal expansivity reach the shallowest depths at low ice shell thicknesses. Above 50 km, simulations with tidal heating enabled reach the shallowest depths due to the significant decrease in conductive lid thickness. The interface with the deepest downwellings is consistently $k(T), \alpha_0$, because of its low convective vigor compared to the other cases. The only two simulations that maintain some high-amplitude variations of the eutectic interfaces (i.e., low frequency signal) are those that employ an ice shell thickness of 60 km and either a variable thermal conductivity ($k(T), \alpha_0$) or a composite rheology (η_{mix}). In the remaining scenarios the eutectic interfaces are dominated by high-frequency signals from the time-dependent destabilization of plumes.

To assess lateral detectability, the depths of these eutectic interfaces need to be compared to the temperature-dependent penetration depths of planetary radar sounders (RIME and REASON), estimated by Kalousová, Schroeder, and Soderlund (2017) to be 100 dB. Figure 3 shows the two-way attenuation fields for simulations with a 40 km ice shell using the low loss model capped at a fixed color once 100 dB is reached. The depth to 100 dB using the high loss model is superimposed on each field as a cyan line.

The temperature dependence of attenuation is clearly demonstrated in Figure 3, as the two-way attenuation field follows the simulation's thermal structure. The two-way attenuation field, along with the 100 dB boundary, reveals similar effects observed in the eutectic interfaces. At high ice shell thicknesses, the simulations with a variable thermal conductivity but constant thermal expansivity ($k(T), \alpha_0$) and those with a composite rheology (η_{mix}) retain the most high-amplitude, low frequency local variations in the conductive lid thickness. On the other hand, the increased convective vigor from simulations with variable thermal expansivity and tidal heating have generally thinner conductive lids with much lower standard deviation of the conductive lid thickness. When comparing all two-way attenuation fields (Supplemental Figure 8 and 9), the strongest variations in maximum penetration depth are in the sluggishly convective simulations, in which downwellings reach substantial depths, therefore increasing the depth to 100 dB by several kilometers.

In order to determine which parameter variations are most critical for the lateral detectability of eutectic interfaces, we compare the depth of each eutectic interface with the depth to 100 dB (for the low or high loss model). Using this comparison, we determine for which simulations and under which thermal conditions the two-way attenuation reaches 100 dB before either the 237.15 K or the 252.05 K eutectic interface is reached.

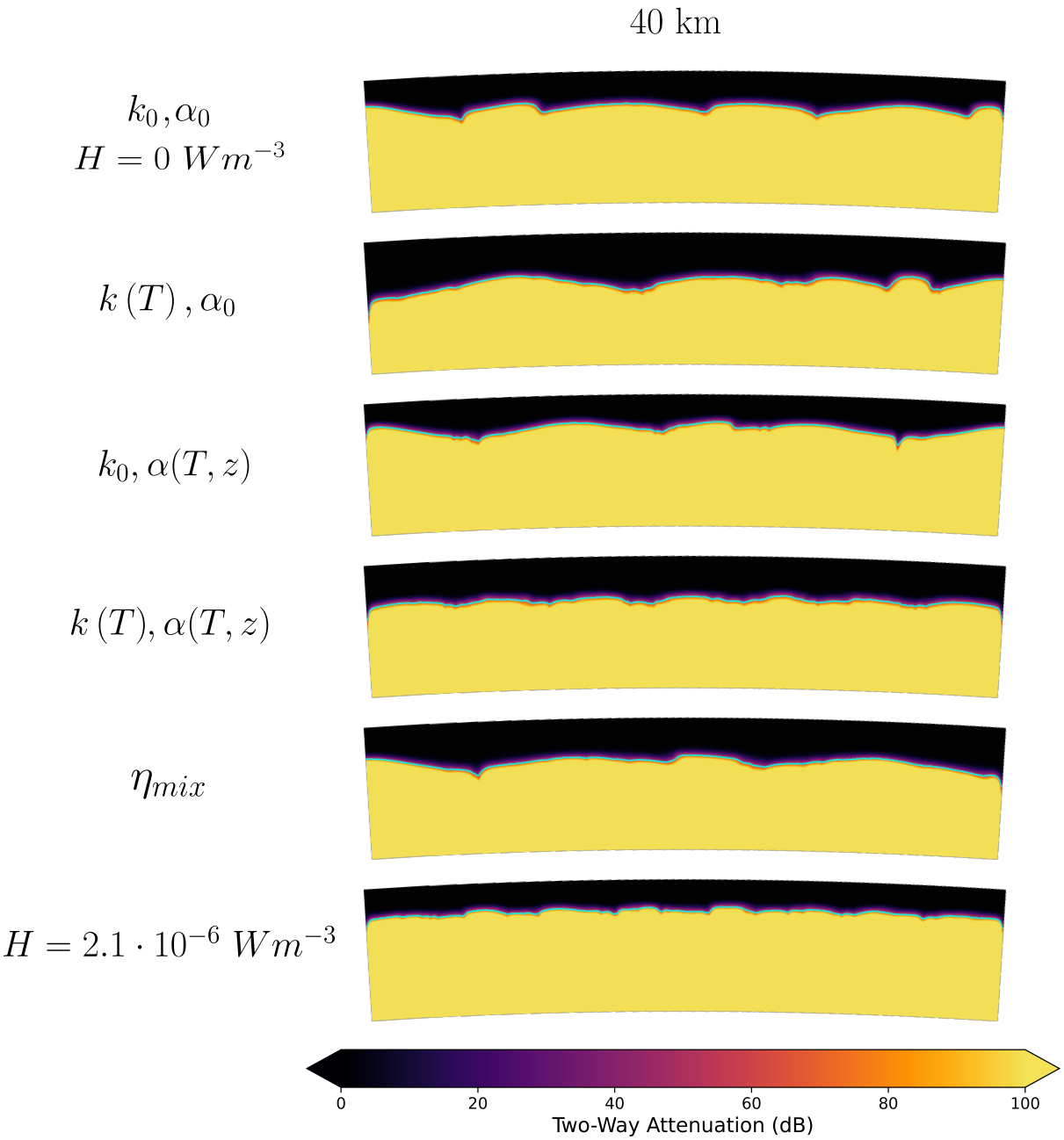


Figure 3: Two-way attenuation fields of non-baseline simulations with a 40 km ice shell, along with the $10^{13}\ Pa \cdot s$ reference simulation. Attenuation field is calculated using the low loss model. Cyan line indicates high-loss 100 dB attenuation interface

Figure 4 illustrates this comparison by plotting each eutectic interface depth along with the 100 dB two-way attenuation depth for various degrees of lateral detectability. All interfaces shown in Figure 4 are from the baseline scenarios. We calculate lateral detectability by combing through each radial column within the simulated domain, determining if 100 dB is reached in that column at some depth prior to a eutectic temperature, and recording the fraction of radial columns that reach this criterion. In the 50 km, 10^{12} Pa · s case (Fig. 4a, 4b), neither eutectic interface reaches a depth below the 100 dB interface for either the low or high-loss models. This results in a lateral detectability of 100% for both eutectic interfaces, since they are always detectable before the 100 dB limit is reached.

For the simulation with a higher reference viscosity of $10^{14}\ Pa \cdot s$, the 237.15 K ($T_{e,Z}$) interface (Fig. 4c) is always laterally detectable, but the depth to the 100 dB high loss interface is occasionally

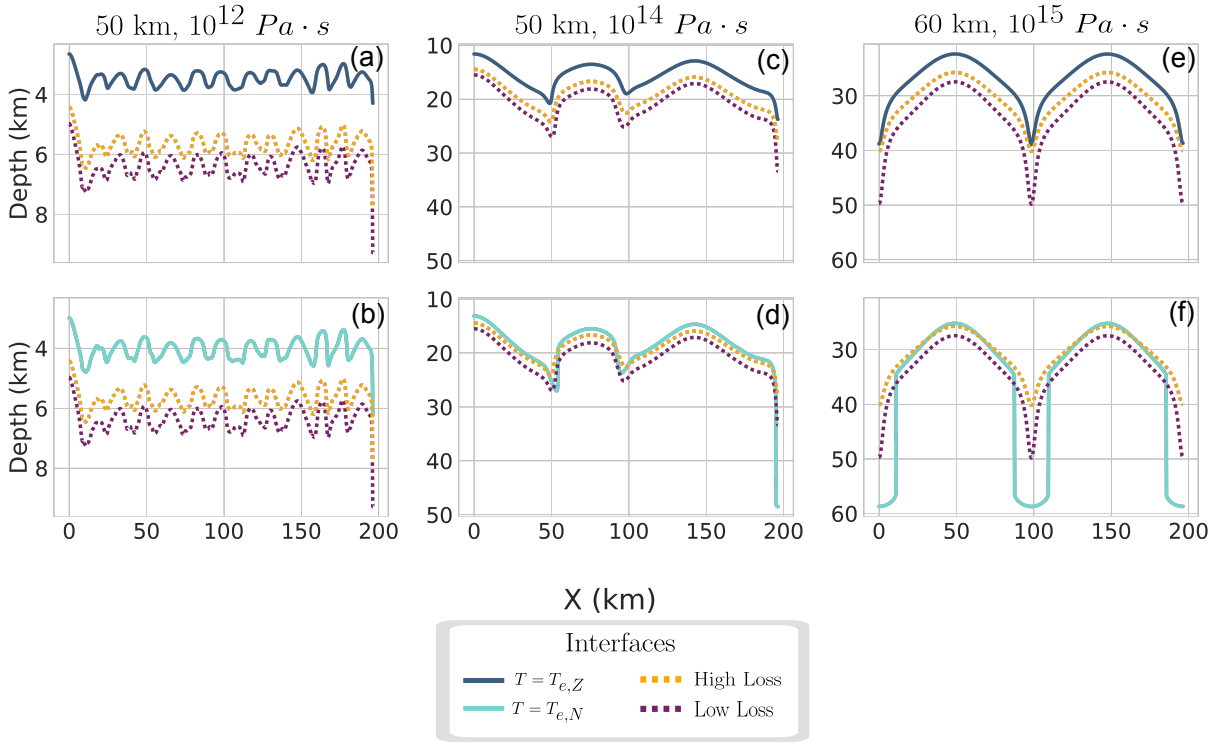


Figure 4: Eutectic interfaces plotted with both low and high loss attenuation interfaces for three simulations of various lateral detectability. Top row (a, c, e) compares the 237.15 K eutectic interface to both attenuation interfaces, while the bottom row (b, d, f) performs the same comparison against the 252.05 K eutectic interface

reached before the higher temperature 252.05 K eutectic interface (Fig. 4d), interestingly at locations of strong downwellings. This simulation has a lateral detectability of 100% with both attenuation models for the 237.15 K eutectic interface, but 98.1% lateral detectability for the 252.05 K eutectic interface with the low loss model and 92.3% lateral detectability with the high loss model.

For the baseline simulations, the lowest lateral detectability found was for the 252.05 K eutectic interface in the 60 km, 10^{15} Pa · s simulation (Fig. 4f), with a lateral detectability of 77.6% for the low loss model and a 34.1% lateral detectability for the high loss model. This significant decrease in lateral detectability is due to the maximum attenuation depth being reached in downwelling regions where the eutectic interface is at its deepest, preventing large chunks of the eutectic interface from being detectable.

Figures 5 and 6 compares lateral delectabilities across all parameter spaces covered by this study. Each heatmap shows lateral detectability for one eutectic interface compared to one attenuation interface (determined by the use of either the low or high loss model) for all baseline simulations (Figure 5) and simulations with more complex mechanisms enabled (Figure 6). Across all simulations (baseline or otherwise), the 237.15 K eutectic interface is 100% laterally detectable for both attenuation models. For the baseline simulations, the 252.05 K eutectic interface is also at 100% lateral detectability simulations with $\eta_{ref} = 10^{15}$ $Pa \cdot s$ and for the two conductive simulations, regardless of attenuation model used. However, there is a significant drop in lateral detectability for the 252.05 K eutectic interface across both attenuation models when crossing into the sluggishly convective regime. When using the low loss model, simulations range between 77.6% to 92.9% lateral detectability for the 252.05 K eutectic interface, while the same simulations vary between 34.1% to 92.2% lateral detectability when using the high loss model. The few examples of vigorously convective simulations that don't reach 100% lateral detectability for the 252.05 K eutectic interface are all simulations with large downwellings, where an attenuation limit is reached just above a downwelling, preventing the

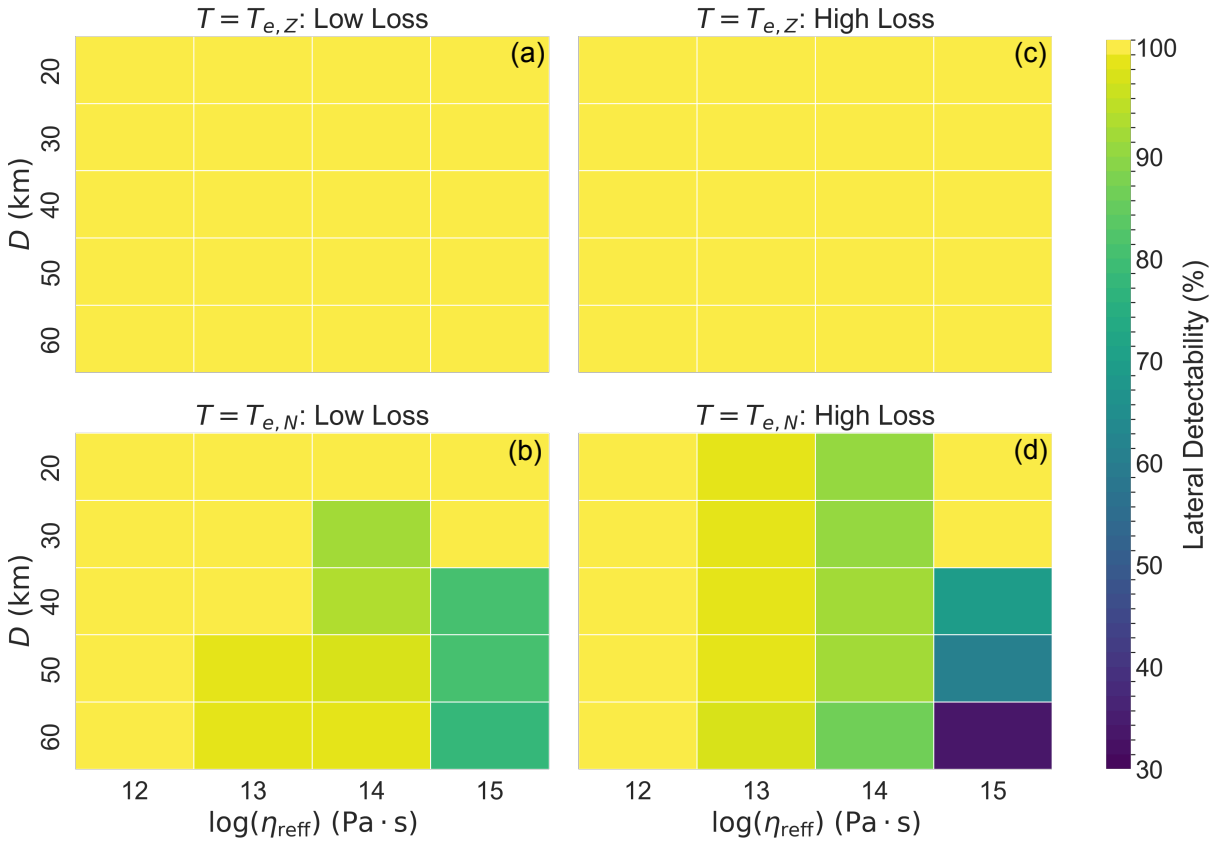


Figure 5: Heatmaps comparing lateral detectability of eutectic interfaces against low (a, b) and high (c, d) loss attenuation models for baseline simulation cases. Eutectic interfaces with 100% detectability are assigned a slightly brighter color for clarity.

detection of the eutectic interface from deep within the ice shell.

Since the non-baseline simulations are all based on a reference viscosity of 10^{13} $Pa{\cdot}s$ (with complex rheology simulations having an equivalent grain size) and therefore will all have high convective vigor, they all have relatively high lateral detectability above 88% (see Fig. 6 color bar). However, the deviations from the baseline $\eta_{ref} = 10^{13}$ $Pa{\cdot}s$ simulations do reveal some interesting trends. To make comparison simpler, we will only compare lateral detectability values for the 252.05 K eutectic interface calculating the depth to 100 dB using the high loss model (Fig. 6d). With these restrictions, baseline $\eta_{ref} = 10^{13}$ $Pa \cdot s$ simulations range in lateral detectability from 98.5% to 99.7%. Simulations with a constant thermal conductivity, but a variable thermal expansivity ($k_0, \alpha(T, z)$), and tidal dissipation all reach lateral detectability values higher than the baseline simulations due to the increased convective vigor.

Both sets of simulations with variable thermal conductivity (with and without a variable thermal expansivity) have lateral detectability values lower than the baseline simulations with high loss and sodium chloride end-member eutectic temperature ($T = T_{e,N}$). In these cases the largest difference with respect to the corresponding baseline case is 9.7% for the 20 km ice shell thickness case with a variable thermal conductivity and constant thermal expansivity ($k(T), \alpha_0$). There does not appear to be a strong correlation between increased ice shell thickness and decreased lateral detectability at reference viscosities below 10^{15} $Pa \cdot s$. There are a few instances where there is a drop in lateral detectability for both the 20 km and 60 km ice shell thicknesses when fixing either reference viscosity or an enabled complex parameter (see $\eta_{ref} = 10^{14}$ $Pa \cdot s$, $k(T), \alpha_0$, and $k(T), \alpha(T, z)$ simulations). We suspect this is due to time-dependent plume destabilization creating extremely deep downwellings, which can occur along the lateral boundaries of the simulation domain (see Fig. 4c, 4d).

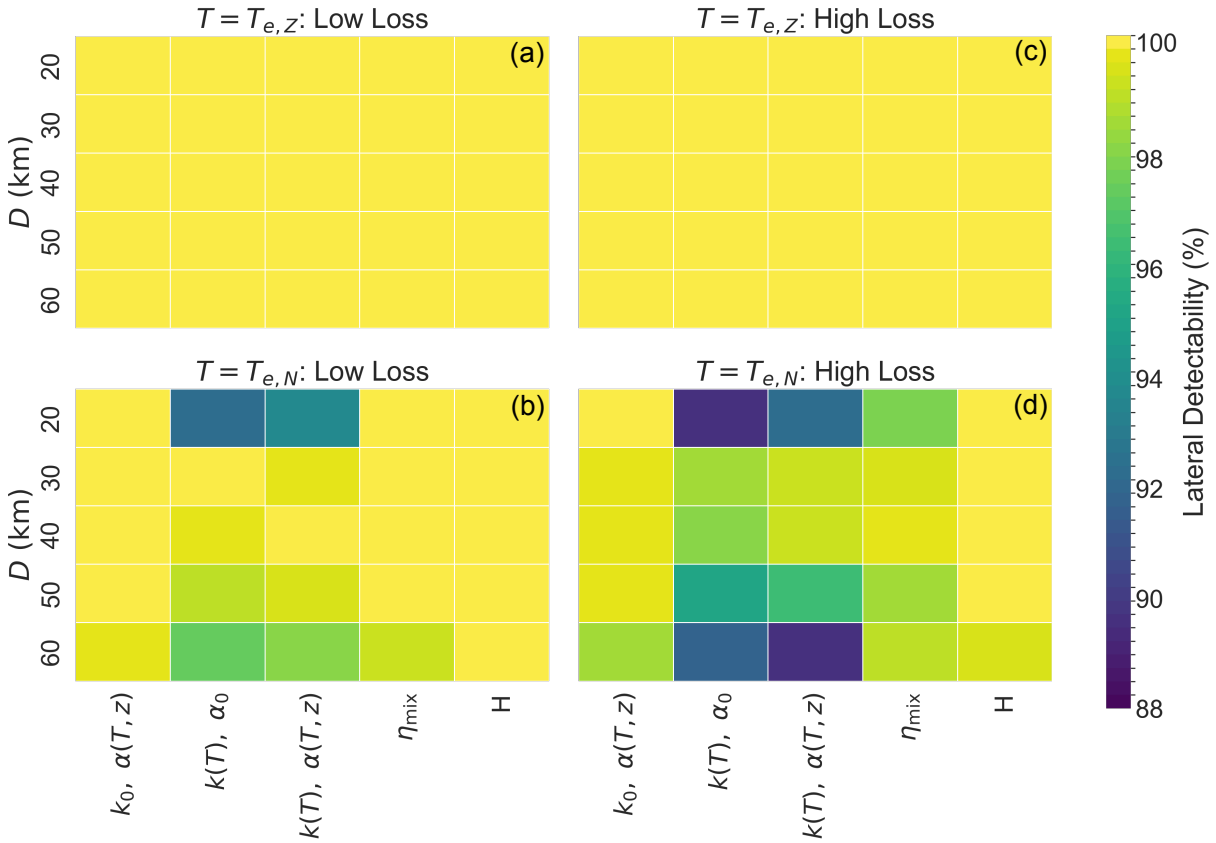


Figure 6: Same as Figure 5 except displays lateral detectability for simulations with variable thermal expansivity, conductivity, complex rheology, and tidal dissipation.

4 Discussion

Our results indicate that the possibility of laterally detecting eutectic interfaces using a planetary radar sounder within Europa's ice shell is high. We find that the 237.15 K eutectic is 100% laterally detectable across all simulations regardless of attenuation model. In a fourth of our cases, the 252.05 K eutectic interface is detectable even in the worst case of the high-loss model.

While many previous studies point out that thermal downwellings serve as a potential source for increased penetration depth for an orbiting radar sounder (McKinnon 2005; Di Paolo et al. 2017; Kalousová, Schroeder, and Soderlund 2017; Souček et al. 2023; DeMers et al. 2025), we have identified a notable decrease in lateral detectability in sluggishly convective simulations. This decrease in lateral detectability is caused by the estimated attenuation limit of 100 dB being reached just above the thermal trough created by downwellings, preventing their deepest portions from being detected.

Recent studies by Kalousová, Schroeder, and Soderlund (2017) and Souček et al. (2023), which investigated the attenuation within the shell of icy moons, have focused on global estimates of sounding penetration depth. This study differentiates itself with its focus on lateral detectability of non-ocean interfaces within the ice shell. The Kalousová, Schroeder, and Soderlund (2017) study of two-way attenuation on Europa's ice shell uses identical parameters for a few of their simulations, so comparing maximum attenuation depths under certain conditions serves as a good benchmark for our geodynamic models. Kalousová, Schroeder, and Soderlund (2017) noted the value of downwellings to increase penetration depth, so they recorded the maximum depth to 100 dB of two-way attenuation achieved using the low-loss model, and the minimum depth achieved using the high-loss model (see Table 2).

All simulations from Kalousová, Schroeder, and Soderlund (2017) were ran with a purely Newtonian viscosity in the same Arrhenius form as equation 14 but within a cartesian box geometry. Their

parameter variations included ice shell thickness, reference viscosity, surface temperature, activation energy and tidal heating. Among these variations, the simulations that share the most similarities with our baseline study have D = 20, 30 km, $\eta_{ref} = 10^{13}$, 10^{14}, 10^{15} $Pa \cdot s$, $T_{surf} = 100$ K, $E_a = 5 \times 10^4$ J/mol, and $H = 0$ W/m^3. We opt to not compare cases with tidal heating, as the H_{max} value selected by Kalousová, Schroeder, and Soderlund (2017) is more representative of a global average and is twice the value we use (5×10^4 W/m^3). When comparing the maximum penetration depth reached in the low loss, our isotherms exist at slightly deeper depths, with a mean difference of 300 meters. Minimum penetration depths reached using the high loss attenuation model are similarly deeper with a mean difference of 230 meters. Kalousová, Schroeder, and Soderlund (2017) also dedicated a small set of simulations to studying how temperature-dependent thermal conductivity modifies ice-ocean detectability. While these simulations use an $\eta_{ref} = 10^{14}$ and slightly different thermal conductivity ranges (2.4 - 5.92 W/mK in this study compared to 2.28 - 5.35 W/mK in Kalousová, Schroeder, and Soderlund (2017)), the low loss case is able to reach the ice-ocean interface in both studies for the 20 km ice shell. This is the only simulation in this study that reaches the ice-ocean interface without first reaching the 100 dB of two-way attenuation, and is also the only simulation that reaches the ice-ocean interface in the set of comparable simulations from Kalousová, Schroeder, and Soderlund (2017). This implies that neither the simulation geometry (either Cartesian box or cylindrical section) nor slight differences in the variable thermal conductivity have a strong effect on the attenuation structure of the ice shell.

There are a few components that this study does not directly address and need to be considered in future work. These components are predominantly complexities not yet included in the current geodynamic model. The interest in eutectic interface stems from the decreased solidus temperature when incorporating salts (observed spectroscopically on the surface (Kargel et al. 2000; Pappalardo, McKinnon, and Khurana 2009)) into an icy matrix, allowing shallow brines to form above 273 K. Like Kalousová, Schroeder, and Soderlund (2017), the dielectric modification of salts is considered in the high-loss model, but salinity variations within the geodynamic model were not. The detection of eutectic interfaces for a radar sounder relies on the dielectric contrast produced from phase distinction between solid ice and liquid water. The density variations of interstitial melt is likely to cause Rayleigh-Taylor instabilities in our geodynamic model, modifying its thermal structure. We did not compute melt generation nor extraction within this model, however previous studies have demonstrated that meltwater drainage to the sub-surface ocean would occur on sub-geologic timescales by either convection (Kalousová et al. 2016) or porous extraction (Kalousová et al. 2014). Since our models solve thermal and not thermo-chemical convection, we are similarly neglecting the effect of salty constituents on general shell dynamics. The addition of a multi-composition geodynamic simulation will modify the depth to compositionally-specific eutectics, and is similarly neglected in this study. Additional modeling efforts are needed to assess the effect of a non-uniform composition on eutectic interface depths.

5 Conclusion

In this study, we showed that the lateral detectability of eutectic interfaces within Europa's ice shell is influenced by the convective regime the shell falls into. While lateral detectability is reached throughout the entire simulated domain for the colder eutectic (i.e., 237.15 K) investigated in this study and associated with a thermodynamic model of Europa's ocean, a warmer eutectic interface (i.e., 252.05 K) corresponding to an end member composition of sodium chloride and pure water ice exhibited reduced detectability for simulations that have just past the critical threshold for buoyancy required to initiate convection. This effect is due to sounding detectability limitations within cold downwellings that places 100 dB two-way attenuation interface at shallower depths than the eutectic interfaces. We observe this behavior across simulations with various degree of complexity, including variable thermal conductivity and expansivity, tidal heating, and the inclusion of multiple deformation mechanisms.

While our investigation shows a decreased lateral detectability for warmer eutectic interfaces under certain convective regimes, the majority of the parameter space varied in this study (both in terms of geophysical and dielectric properties) demonstrates a strong likelyhood of lateral eutectic detectability being achieved. We do not observe a lateral detectability percentage below 30% in any or the simulations. However, these results are tied very closely to the thermal structure of our geodynamic model. Current geodynmaic modeling of attenuation within icy moons is limited to purely thermal convection, and neglect the effect of a chemically heterogeneous shell on. Future studies should observe the effect of a varied composition on eutectic interface depths, and attempt to incorporate more realistic behaviors (e.g. sparse echoes) observed in terrestrial radagrams of glacial structures to more accurately provide detectability limitations, both laterally and radially.

Open science declarations

Author contributions. W.B. is responsible for conceptualizing, data curation, formal analysis, investigation, software, visualization, and writing (original draft). A.P. was responsible for funding, project administration, and resources. All authors were additionally responsible for supervision, validation, and writing (review & editing).

Code availability. All code used in this study is available at https://github.com/AtWillB/gaia_eutectic_constrain. The code base used to gather data from the simulations can be found under the src folder, while code used to generate figures for this study can be found under data/documentation/thesis/figures/paper.

Data availability. The following Zenodo repository is a clone of the github repository used for this project, and serves to fulfill the data availability requirement: https://doi.org/10.5281/zenodo.20367706. Data used to generate all figures can be found under the data/axillary folder inside the repository.

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Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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